

Episcaf: project log

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June 17, 2026

Living record. Read top to bottom to get up to speed. Figures regenerated from `results/`; see `docs/WORKFLOW.md`

This is the running record of the episcaf project: scaffolding antibody epitopes defined by 3D geometry into folded, assay-ready designs (RFDiffusion3 $\rightarrow$ ProteinMPNN $\rightarrow$ AlphaFold3). It is meant to be read in order, since each section is one iteration: what we tried, what we found, and what it changed for the next step. It is not a formal manuscript. Every figure is regenerated by a named script (recorded in `manuscript/figures/FIGURES.md`) and every number traces to `results/`; unconfirmed values are flagged [verify].

1 Motivation: scaffolding conformational epitopes

1.1 Why scaffold epitopes

Antibodies recognize their antigens through epitopes defined by three-dimensional geometry. Many of the most interesting epitopes are discontinuous in sequence but contiguous in space: the residues an antibody contacts are scattered along the antigen sequence yet fold together into a single surface patch. This is precisely the kind of target that linear methods cannot represent. High-throughput serology platforms such as PepSeq tile an antigen into short, overlapping linear peptides, which works well for continuous, sequence-local epitopes. A linear peptide, however, cannot reconstitute a surface assembled from sequence-distant residues, and it discards the native conformation the antibody evolved to recognize, so conformational and discontinuous epitopes fall through the gaps of linear tiling.

The aim of this project (“episcaf”) is to close that gap. We take an antibody epitope defined by its native three-dimensional geometry and build a folded, assay-ready protein that displays the epitope the way it sits on the antigen. The longer-term goal is to extend PepSeq-style high-throughput assays beyond linear peptides to conformational epitopes.

1.2 The approach

We extract the epitope, meaning the residues an antibody contacts, from an antibody:antigen crystal structure. We embed those residues into a *de novo* protein scaffold with RFDiffusion, assign a sequence with ProteinMPNN, and then ask AlphaFold3 whether the design actually folds with the epitope held in its native geometry. Designs are kept or rejected on AlphaFold3 confidence and geometry-preservation metrics (Section 2). The full production pipeline, including the Nextflow orchestration, the RFDiffusion runs, and the AlphaFold3 evaluation, originates in Lawson Woods’ `episcaf-experiments` repository. Reproducing and then extending it is the starting point of this work.

1.3 Patches and islands

A single antibody:antigen complex can yield more than one epitope *patch*, labeled 0P, 1P, and so on, where each patch is a distinct cluster of antibody-contacting residues drawn from the same structure. A patch is in turn made of one or more *islands*, the spatially contiguous runs of epitope residues in three-dimensional space. This distinction matters because it predicts difficulty. Single-island patches can be enforced locally and tend to scaffold well, whereas multi-island patches require preserving the relative position and orientation of spatially separated regions, which is much harder.

A concrete failure makes the point. PDB 2XQB, patch 0P, is a 56-residue epitope split across two separated islands. Across 2,624 RFdiffusion trials, zero designs passed the downstream structural filters. Because this 0% success rate persists over thousands of attempts, it reads as a robust computational failure rather than undersampling. The cause appears procedural: contigs (the constraints RFdiffusion is given) enforce epitope residue identity and ordering, but they do not strongly constrain the distances and orientations *between* islands, so the model can satisfy the contig and still lose the geometry an antibody needs. Single-island, compact patches run through the identical pipeline and filters do produce passing designs, and the `designFilterRates` summary shows success rates that correlate strongly with epitope topology, particularly the number of islands.

1.4 Two regimes: with and without a known antibody

The project runs in two settings that differ in what “correct” can be measured against.

- **Known-antibody (the “DP3” set).** When the antigen has a solved antibody complex, we have a ground truth: a design is good if its epitope is held in place and the scaffold does not collide with where the antibody must bind. This gives the four-filter pass definition used throughout (Section 2).
- **No-antibody (the tiled 12-mer set).** Most targets have no solved antibody, so there is nothing to clash against. Here we tile the antigen into 12-residue epitopes and replace the antibody-clash filter with a native-antigen-aware accessibility surrogate (Section 2) that asks whether scaffold mass intrudes into the volume an approaching antibody would need.

A single, config-driven composite scorer ranks designs in both regimes, differing only in which accessibility term it uses.

1.5 How the rest of this log reads

The sections that follow are in the order the work happened. We start by reproducing Lawson’s workflow and confirming that our reimplementations of the metrics matches his (Section 2). We then turn to the filtering, where the original global metrics bias selection toward rigid, well-ordered folds, and show how decomposing them into epitope- and scaffold-specific terms widens the designable space (Section 3). From there we build a composite score that selects the best designs per epitope (Section 4), and finally extend everything to antigens with no known antibody using the native-aware cylinder surrogate (Section 5). Each step motivates the next.

2 The pipeline, and reproducing Lawson’s workflow

2.1 Pipeline

RFdiffusion3 generates scaffold backbones around the epitope, ProteinMPNN re-sequences them, and AlphaFold3 predicts the structure single-sequence (`--norun_data_pipeline`, one diffusion

sample, seed 1). Per-design metrics are computed from the AlphaFold3 prediction against the designed structure. Generation code lives in `episcaf_pipeline/`, and the metrics, scoring, and visualization code in `episcaf_analysis/`.

2.2 Reproducing Lawson’s baseline

Before trusting any new number we re-implement Lawson’s metrics and check them against his ground truth. `compute_metrics.py --validate` reads `dp2.parquet` (Lawson’s stored per-design values), recomputes `overall_rmsd` and `epitope_chunk_rmsd` from his ProteinMPNN PDBs and AlphaFold3 CIFs, and diffs them against the stored values. Agreement on this diff is the precondition for everything downstream: it confirms that our implementation of the filters matches the workflow we are extending, so that later differences reflect the design protocol rather than a scoring bug.

2.3 Four-filter ground truth (DP3 set)

A design passes when all four conditions hold: overall backbone RMSD ≤ 2 Å, epitope-chunk RMSD ≤ 1 Å, mean predicted aligned error < 5 , and zero antibody residues within 4 Å of non-epitope scaffold residues (`af3_n_clash_res`). Clash detection Kabsch-aligns the AlphaFold3 epitope onto the true epitope, then checks non-epitope scaffold heavy atoms against antibody atoms. The antibody coordinates always come from the AbDb crystal structure, not from AlphaFold3.

2.4 Decomposed metrics

The two continuous filters, overall RMSD and mean PAE, treat the epitope and the scaffold as a single unit. We split each into epitope- and scaffold-specific terms (epitope RMSD, scaffold RMSD, epitope PAE, scaffold PAE), retaining the zero-clash hard filter throughout. Decomposition lets selection reward a well-placed epitope on a flexible scaffold (or the reverse) instead of demanding global rigidity, and it lets us ask whether the optimal rigidity profile is epitope-dependent rather than universal (Section 3). Selection over the four metrics is organized as a 2×2 of strict and relaxed thresholds on (epitope, scaffold), with a flexible/flexible arm as a negative control.

2.5 Native-antigen-aware cylinder surrogate (12-mer set)

With no antibody available, we approximate the occluded volume with a cylinder fitted to the epitope. The epitope C α atoms define a plane by SVD, and the outward normal (the antibody approach direction) orients a cylinder of radius 16 Å, height 40 Å, and offset -4 Å. We count the non-epitope scaffold C α atoms inside the cylinder. The native-aware variant excludes scaffold atoms within a short distance of a native-antigen heavy atom, i.e. residues sitting where the native antigen already sits, which a real antibody tolerates. The geometry core is `episcaf_analysis/native_cylinder_core.py` (self-tested).

2.6 Composite scorer

One config-driven scorer (`episcaf_analysis/score.py`) ranks designs in four steps: (i) **gate** on epitope-chunk RMSD; (ii) transform each metric to a higher-is-better score within a scope (pooled, or per antigen); (iii) **weighted sum**; and (iv) keep the **top- k per epitope**. Each metric is one input with its own activation (percentile, min-max, z-score, or sigmoid), so the model is effectively

a single-layer perceptron with hand-set weights, feed-forward and without backpropagation. The dials live in `episcaf_analysis/presets.py`:

metric	weight
epitope-chunk RMSD	0.35
epitope PAE	0.25
accessibility	0.25
overall RMSD	0.15

The weights sum to one (renormalized over whichever metrics a given table actually contains). The only term that changes between the two regimes is accessibility: it is `cylinder_native_aware` for the no-antibody 12-mer set and `cylinder_ca_clashes` for the DP3 set, using the same cylinder geometry, with the native-aware carve applied only where there is no real antibody to clash against. The 12-mer preset transforms metrics per antigen and keeps the top 5 designs per epitope, while the DP3 preset transforms pooled and keeps the top 15. The weights are a hand-set starting point; Section 6 describes fitting them to the DP3 `is_pass` label.

3 Rethinking the filters: global metrics bias toward rigid folds

With the pipeline reproduced and trusted (Section 2), the first thing that needed rethinking was the filtering. The original filters score a design with two global continuous metrics, overall RMSD and mean PAE, that treat the epitope and the scaffold as one unit. This is a reasonable inheritance from standard de novo practice rather than an arbitrary choice, but it biases selection toward globally rigid, well-ordered folds and silently rejects designs whose scaffold is flexible while the epitope is well positioned (or the reverse). Decomposing each metric into epitope- and scaffold-specific terms (Section 2.4) is what lets us see, and eventually recover, that rejected-but-viable space.

To compare design sets on a common footing we use epitope-chunk RMSD $< 2.5 \text{ \AA}$ as a single bar. All three tiled 12-mer antigens pass it more often than the pooled DP3 set:

set	pass rate ($< 2.5 \text{ \AA}$)	median epitope RMSD ($\text{\AA}$)
DP3 / mAb (pooled)	30.0%	5.00
6M0J	47.0%	2.65
1D2K	60.2%	1.85
4WAT	84.4%	0.43

Even the hardest 12-mer antigen (6M0J) clears the bar more often than DP3. The full six-metric distribution comparison is Fig. 1.

Caveats (important for interpretation). This comparison is not yet apples-to-apples. The 12-mer epitope is a single contiguous 12-residue stretch, whereas DP3 epitopes are larger and often multi-island, so 12-mer RMSD/PAE are mechanically lower in part by definition, and the pooled DP3 30% also averages over heterogeneous epitopes. The clean control is to stratify within DP3 by island count and size (Section 6). Cylinder counts are likewise not comparable across sets. The 12-mer cylinder medians (~ 31 – 32) exceed DP3 (~ 21) mainly because the longer scaffolds place more $\text{C}\alpha$ atoms in a fixed cylinder, a scaffold-size artifact rather than more clashing. As a sanity check, the DP3 cylinder median (≈ 21) tracks its real `af3_n_clash_res` median (≈ 19).

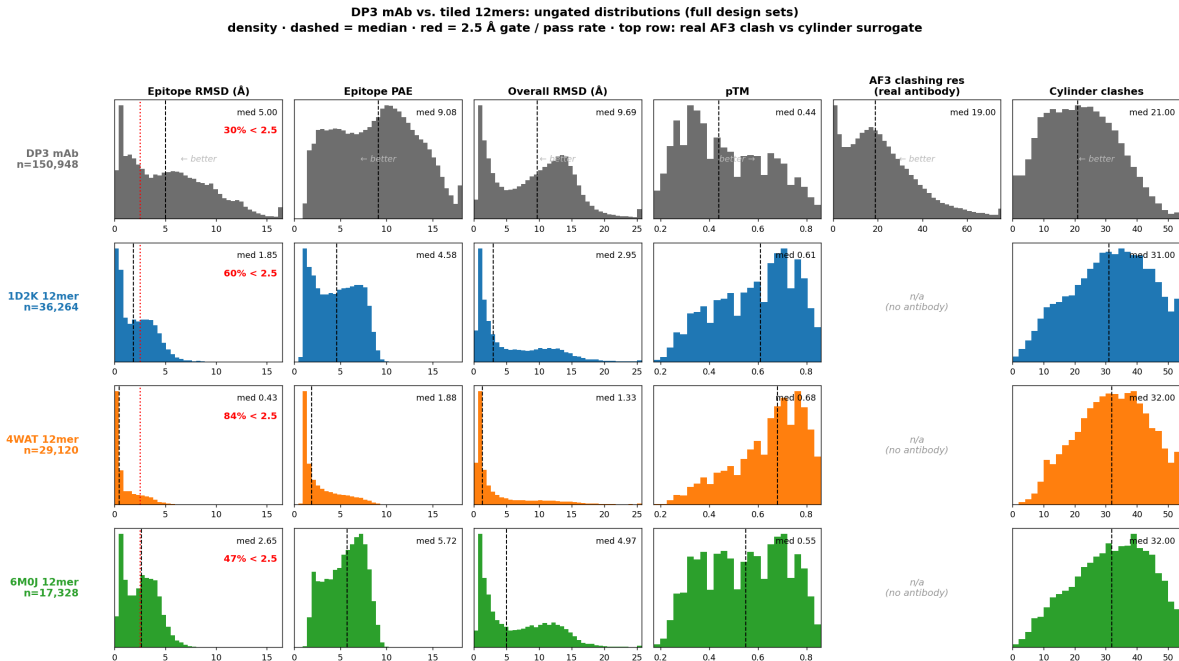


Figure 1: Ungated metric distributions (full design sets), DP3 mAb versus the three 12-mer antigens, across epitope RMSD, epitope PAE, overall RMSD, pTM, AF3 clashing residues (antibody set only), and cylinder clashes. Dashed line marks the median; red marks the 2.5 Å epitope-RMSD gate with the per-set pass rate. Regenerated by `notebooks/dp3_vs_12mer_distributions.ipynb`.

4 Composite scoring and per-epitope selection

Decomposed metrics give more knobs, which raised the next question: how to combine them into a single ranking so we can take the best designs per epitope. The answer is the config-driven composite scorer (Section 2). It gates on epitope-chunk RMSD, transforms each metric to a higher-is-better score within its scope (per antigen for the 12-mer set, pooled for DP3), weight-sums, and keeps the top designs per epitope. Run over the full 12-mer set, the gate keeps 54 539 of 82 712 designs ($\approx 65.9\%$ pooled) before ranking [regen], consistent with the per-antigen rates above.

The scorer is deliberately simple: one metric per input, each with its own activation, weighted and summed, which makes it a single-layer perceptron with hand-set weights. It is still in flux. The per-metric transform is currently a percentile within the population, so a design’s score depends on the cohort it is ranked against. Whether to switch to a population-independent sigmoid (an absolute, saturating per-metric score) is an open choice carried in Section 6.

5 Scaffolding without an antibody: the cylinder surrogate

Most targets of interest have no solved antibody, so the four-filter ground truth does not apply, because there is no antibody to clash against. We extend the workflow to these antigens by tiling them into 12-residue epitopes and replacing the antibody-clash filter with a native-antigen-aware cylinder surrogate (Section 2): a cylinder fitted to the epitope plane along the antibody-approach normal that counts non-epitope scaffold mass intruding into the volume an approaching antibody would need (Fig. 2).

A naive cylinder over-penalizes, because some scaffold mass sits exactly where the native antigen

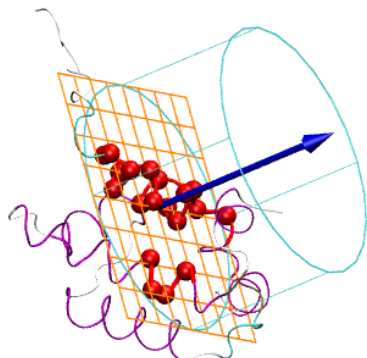


Figure 2: The accessibility cylinder. Epitope residues (red) sit on the scaffold (magenta); the epitope C α atoms define a plane (orange grid) whose outward normal (blue arrow, the antibody-approach direction) orients the cylinder (cyan). Scaffold C α atoms inside the cylinder count against accessibility. Rendered in VMD by `known_antigen/analysis/cylinder_vis/visualize_cylinder.tcl`.

itself already sits, a volume a real antibody clearly tolerates since it binds the native antigen there. The native-aware carve fixes this: among truly clash-free designs, we exclude scaffold C α atoms within 1 Å of a native-antigen heavy atom. Concretely, of the 3186 DP3 designs with zero true antibody clash, the number the plain cylinder flags as heavily penalized (count ≥ 10) falls from 1112 to 384 once the native footprint is carved out, which is **728 false-positive penalties removed**, while genuinely clashing designs are left essentially untouched (Figs. 3 and 4).

A pipeline constraint worth recording: crystal gaps. Epitope fidelity for the 12-mer set is measured by aligning the AlphaFold3 epitope onto the native crystal epitope, matched by residue number (`build_12mer_metrics.py`). When a tile’s epitope residues are unresolved in the crystal, meaning a gap in the deposited structure, the native C α anchors are missing, the Kabsch alignment cannot be formed, and the tile is dropped with status `crystal_epi_incomplete`. This is a real constraint on which 12-mer epitopes are scorable at all: an antigen whose crystal has gaps over the tiled region contributes fewer (or no) scorable tiles, independent of design quality. The mechanism is confirmed in code; the specific antigen(s) and tile counts affected are not in the local ok-only export and remain to be pulled from the cluster pre-filter status counts [`verify`].

6 Open questions and next steps

1. **Within-DP3 island stratification (the clean control).** Split the DP3 set by epitope island count and size and recompute pass rates, to separate the “contiguous-12 is easier” confound from a real scaffolding-difficulty effect. This needs the per-epitope island metadata in `datasets/dp2.parquet`.

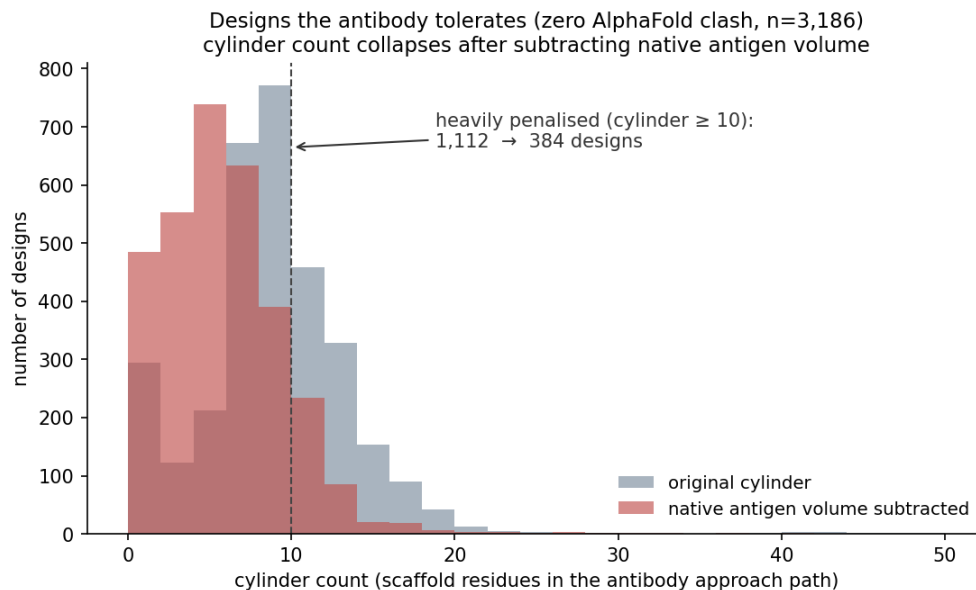


Figure 3: Cylinder count among clash-free DP3 designs (zero AlphaFold3 antibody clash, $n = 3186$), before (grey) and after (red) subtracting the native-antigen volume. The heavily penalized tail (count ≥ 10) collapses from 1112 to 384. Regenerated by `episcaf_analysis/viz/plot_fp_reduction.py --native metrics_native_cyl.csv --mode dist`.

2. **Per-metric transform: percentile/rank versus sigmoid (undecided).** The composite currently maps each metric to a percentile within its population, so a design’s score depends on the cohort it is ranked against and is not comparable across runs. A sigmoid transform (with a fixed midpoint and slope k per metric) would instead give a population-independent, absolute saturating score, answering “how good is this overall RMSD on its own merits” rather than “relative to this batch”. The scorer already supports both (`transform=sigmoid` in `presets.py`); the open question is which should be the default, and how to set each metric’s midpoint and k . I am leaving this unresolved until we see whether absolute scores transfer better across antigens.
3. **Learn the scorer weights.** The DP3 set carries the four-filter `is_pass` label. Fit a logistic regression of the transformed metrics on `is_pass` to replace the hand-set weights, then transfer them to the 12-mer preset where no label exists. This and the transform choice above are the two halves of turning the hand-set perceptron into a fitted one.
4. **Confirm the canonical DP3 metrics builder.** Determine whether `compute_metrics_mpn.py` (currently archived) is the live RFD3+MPNN metrics producer, and promote it into `episcaf_analysis/` if so (see `docs/MIGRATION.md`).
5. **Validate the DP3 scoring path** end-to-end with `--preset antibody`, confirming `cylinder_ca_clashes` is the correct accessibility column there.
6. **Finish the 12-mer AlphaFold3 data migration** and verify landing counts against the manifest.

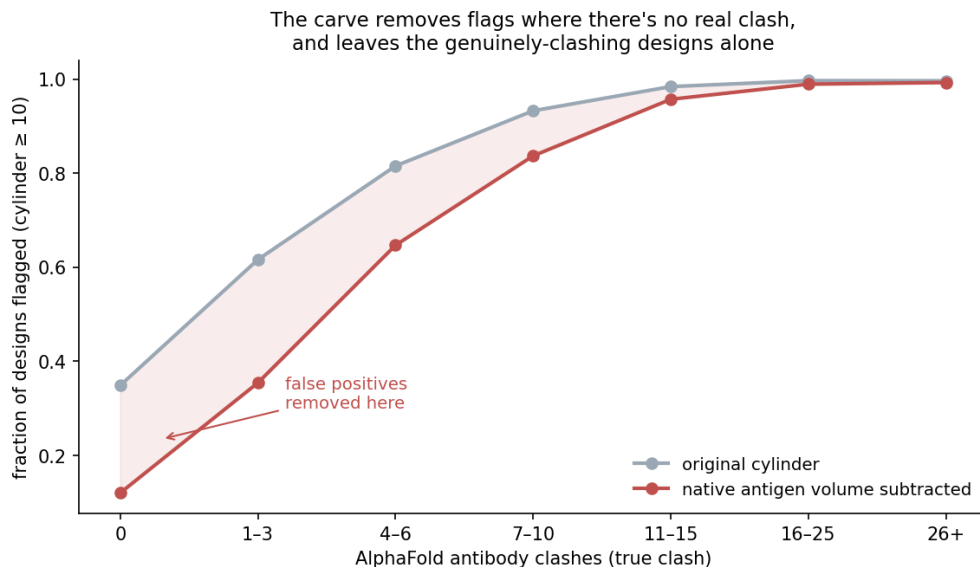


Figure 4: The carve removes flags selectively. Fraction of designs flagged (cylinder ≥ 10) across bins of true antibody clash, before versus after the carve. The gap is large where there is no real clash (false positives removed) and closes where the clash is real (true signal retained). `--mode selectivity`.

7. **Quantify crystal-gap dropouts.** Pull the cluster pre-filter status counts to report, per antigen, how many tiles were dropped as `crystal_epi_incomplete` (epitope residues unresolved in the deposited crystal; Section 5). This turns the recorded mechanism into a verified number, and tells us whether any intended antigen was effectively un-scorable from gaps alone.

References